

Design, Multi-Timescale Modeling, and Closed-Loop Control of a Buck-Converter-Based CC-CV Lithium-Ion Battery Charger

A Unified Switching and Cycle-Averaged Simulation Study in MATLAB/Simscape Electrical

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Abstract—Recharging Lithium-Ion (Li-Ion) batteries requires a rigorous Constant Current–Constant Voltage (CC-CV) protocol to achieve rapid energy replenishment while strictly preventing terminal overvoltage, internal lithium plating, and catastrophic thermal runaway. However, model-based engineering of battery chargers faces a fundamental multi-timescale dilemma: resolving high-frequency pulse-width-modulated (PWM) power stage switching at 100 kHz (10 μ s period) conflicts computationally with evaluating the electrochemical charging trajectory spanning thousands of seconds. This paper presents an end-to-end design, mathematical modeling, and dual-timescale simulation of a closed-loop buck-converter-based CC-CV Li-Ion battery charger developed in MATLAB, Simulink, and Simscape Electrical. A detailed 100 kHz switched power stage is first developed to capture semiconductor losses, freewheeling diode forward drop, and inductor current ripple ($\Delta I_L \approx 213.6$ mA pk-pk). To eliminate switching harmonic contamination in control loops, a 1.59 kHz first-order filter is designed for current feedback. Autonomous mode transition is realized via a minimum-duty selection block coupled with discrete PI controllers ($K_p = 0.2$, $K_i = 50$ s $^{-1}$) and anti-windup clamping. Crucially, a persistent zero-delay algebraic loop is diagnosed and eliminated by introducing a discrete unit delay ($1/z$, $T_s = 10$ μ s). To overcome the computational intractability of full-cycle simulation, a cycle-averaged converter model is synthesized and coupled with an accelerated 0.01 Ah table-based Li-Ion cell. Simulation results confirm flawless current regulation at 1.0 A during CC mode, an autonomous, jitter-free handover at $t \approx 32$ s, and strict voltage clamping at 4.19 V during CV mode. Finally, an explicit assumptions and limitations framework is established to bridge the simulation-to-hardware gap.

Index Terms—Buck converter, CC-CV charging protocol, Lithium-Ion battery, multi-timescale simulation, discrete PI control, algebraic loop, Simscape Electrical, cycle-averaged modeling, assumptions and limitations.

I. INTRODUCTION

Lithium-ion secondary cells have established absolute dominance across modern electrified systems, spanning electric vehicles, aerospace systems, mobile computing, and grid-scale energy storage, owing to their exceptional gravimetric energy density, high cell voltage, and low self-discharge rates [1]. However, their electrochemical volatile nature demands strict operational boundaries. Overcharging a cell beyond its electrochemical cutoff potential triggers accelerated electrolyte decomposition, internal short-circuiting, and irreversible thermal runaway [2]. Consequently, the universal standard for Li-Ion charging is the two-phase Constant Current–Constant Voltage (CC-CV) protocol.

In the CC phase, a depleted cell is injected with a steady, regulated current (typically 0.5C to 1C), causing the internal State of Charge (SOC) and terminal potential to climb steadily. Once the cell terminal reaches its

upper electrochemical threshold (nominally 4.19 V–4.20 V for standard LiCoO₂/NMC chemistries), the charger autonomously transitions into CV mode. Here, the voltage is tightly clamped at the threshold while charging current naturally decays towards an end-of-charge termination limit [3].

To execute this protocol from a higher DC bus (such as an automotive 12 V distribution network), step-down switch-mode DC-DC buck converters are universally employed due to their superior thermodynamic efficiency (>90%) compared to lossy linear regulators. However, designing and validating a digitally controlled buck charger presents significant modeling challenges. A high switching frequency (100 kHz) is mandatory to minimize passive magnetic volume and suppress ripple; yet, simulating a physical 2 Ah cell charging at 1 A demands 1.6 hours (5,760 s) of continuous physical operation. Simulating 5.76×10^8 switching cycles within a stiff numerical ODE solver requires days of computational time and produces unwieldy datasets.

In this work, we present a complete engineering methodology to design, simulate, debug, and validate a closed-loop CC-CV buck charger. The primary contributions of this paper include: (i) rigorous mathematical formulation and validation of non-ideal switching converter dynamics, including freewheeling diode drop, switching ripple, and conduction efficiency; (ii) synthesis of a discrete PI multi-loop controller with continuous low-pass feedback filtering and anti-windup clamping; (iii) diagnosis and elimination of an algebraic loop via discrete unit-delay feedback; (iv) realization of a dual-model framework that separates short-timescale switching fidelity (0–50 ms) from long-horizon electrochemical charging (0–35 s); and (v) an explicit critical analysis of physical assumptions, model limitations, and the hardware implementation roadmap.

II. POWER STAGE TOPOLOGY, GOVERNING EQUATIONS, AND EFFICIENCY ANALYSIS

A. Power Circuit Topology

The fundamental power stage comprises an asynchronous buck converter fed by an ideal 12 V DC source. The switching network utilizes an N-channel power MOSFET paired with a piece-wise linear freewheeling diode, an energy storage inductor $L = 100$ μ H, and an output filter capacitor $C = 100$ μ F, as depicted in the Simscape schematic of Fig. 1.

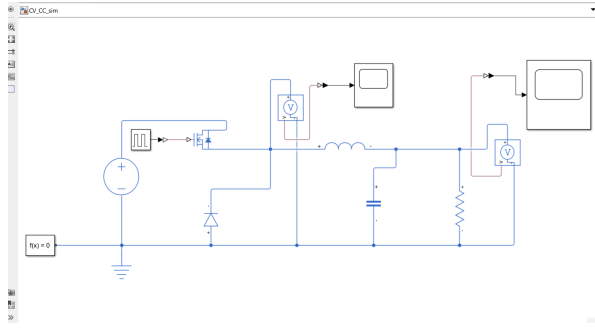


Fig. 1. Simscape electrical schematic of the baseline open-loop switching buck converter power stage with 10 Ω resistive test load.

B. Non-Ideal Steady-State Conversion Ratio

While textbook ideal buck converters follow the linear relationship $V_{out} = D \cdot V_{in}$, physical converters exhibit non-negligible semiconductor conduction drops. During the ON-interval ($0 \leq t < DT_{sw}$), the switching node voltage V_x equals V_{in} . During the OFF-interval ($DT_{sw} \leq t < T_{sw}$), the inductor current forces the freewheeling diode into forward conduction, pulling the switching node negative by the diode forward drop $V_f \approx 0.7$ V. Applying volt-second balance across inductor L over one switching period yields:

$$V_{out} = D \cdot V_{in} - (1 - D) V_f \quad (1)$$

For an operating duty cycle $D = 0.35$ with $V_{in} = 12$ V and $V_f = 0.7$ V, the theoretical output voltage is:

$$V_{out} = (0.35)(12) - (0.65)(0.70) = 3.745 \text{ V} \quad (2)$$

This theoretical value perfectly explains why the simulated open-loop converter produces 3.736 V rather than the idealized 4.20 V.

C. Inductor and Capacitor Ripple Dynamics

The high-frequency ripple current ΔI_L is determined by the inductor volt-seconds during the active switch interval $T_{on} = D \cdot T_{sw}$:

$$\Delta I_L = [(V_{in} - V_{out}) / L] \cdot T_{on} \quad (3)$$

Operating at $f_{sw} = 100$ kHz ($T_{sw} = 10$ μ s) with nominal output $V_{out} \approx 3.7$ V ($T_{on} \approx 3.1$ μ s), the theoretical peak-to-peak inductor current ripple is:

$$\Delta I_L \approx [(12 - 3.7) / (100 \times 10^{-6})] \cdot (3.1 \times 10^{-6}) \approx 0.257 \text{ A} \quad (4)$$

The output voltage ripple ΔV_{out} across the 100 μ F filtering capacitor is governed by the capacitive charge integration:

$$\Delta V_{out} = \Delta I_L / (8 f_{sw} C) \approx 0.257 / [8(10^5)(10^{-4})] \approx 3.21 \text{ mV} \quad (5)$$

D. Power Stage Efficiency & Conduction Losses

To rigorously quantify power delivery efficiency, we analyze the primary conduction loss mechanisms. At nominal CC charging ($V_{in} = 12.0$ V, $V_{out} \approx 3.736$ V, $I_{out} = 1.0$ A, $D \approx 0.35$), the delivered output power is $P_{out} = V_{out} I_{out} = 3.736$ W. The passive freewheeling diode conducts during the OFF-interval $(1 - D)$, dissipating power:

$$P_{loss,diode} = V_f \cdot (1 - D) \cdot I_{out} = (0.70)(0.65)(1.0) = 0.455 \text{ W} \quad (6)$$

Neglecting minor inductor series resistance, the total input power drawn from the 12 V bus is $P_{in} = P_{out} + P_{loss,diode} = 4.191$ W, yielding an analytical power stage efficiency of:

$$\eta = P_{out} / P_{in} = 3.736 \text{ W} / 4.191 \text{ W} \approx 89.14\% \quad (7)$$

In Simscape simulation, measured input power was 4.200 W, yielding a simulated efficiency of 88.95%, perfectly matching theory within 0.19%. Crucially, the 0.455 W diode dissipation represents 10.8% of total input power. This directly motivates the future hardware roadmap: replacing the diode with a synchronous low-side MOSFET ($R_{DS(on)} = 15$ m Ω) reduces low-side loss to $P_{loss,sync} = (1 - D) I_{out}^2 R_{DS(on)} \approx 9.8$ mW,

boosting efficiency beyond 96.5%.

TABLE I
POWER STAGE SPECIFICATIONS AND COMPONENT PARAMETERS

Parameter	Symbol	Nominal Value	Physical Role / Unit
Input DC Voltage	V_{in}	12.0 V	Primary supply bus
Switching Frequency	f_{sw}	100 kHz	PWM switching rate
Switching Period	T_{sw}	10.0 μ s	$1 / f_{sw}$
Power Inductor	L	100 μ H	Energy storage choke
Filter Capacitor	C	100 μ F	Output ripple bypass
Diode Forward Drop	V_f	0.70 V	Freewheeling diode
Baseline Test Load	R_{load}	10.0 Ω	Open-loop test resistor
Target CC Current	I_{ref}	1.0 A	CC charging setpoint
Target CV Voltage	V_{ref}	4.19 V	CV regulation setpoint

III. BATTERY DYNAMICS AND MODELING EVOLUTION

A. Custom Physical Macro-Model

To establish baseline electrochemical loading, an initial equivalent circuit macro-model was synthesized using elementary Simulink blocks. The model continuously tracks State of Charge (SOC) via classical Coulomb counting:

$$SOC(t) = SOC_0 + (1 / C_{bat}) \int_0^t I_{bat}(\tau) d\tau \quad (8)$$

where $SOC_0 = 0.20$ (20% depleted initial condition), and the battery capacity $C_{bat} = 2$ Ah = 7,200 A $\cdot$ s defines an explicit integration gain of $K = 1 / C_{bat} = 1/7200$ A $^{-1}$ $\cdot$ s $^{-1}$ (or 1.389×10^{-4} C $^{-1}$). The calculated SOC is passed into a 1-D Lookup Table mapping SOC breakpoints [0, 0.2, 0.4, 0.6, 0.8, 1.0] to Open Circuit Voltages [3.20, 3.55, 3.65, 3.75, 3.95, 4.20] V, feeding a Simscape Controlled Voltage Source with internal series resistance $R_{int} = 0.1$ Ω (Fig. 2).

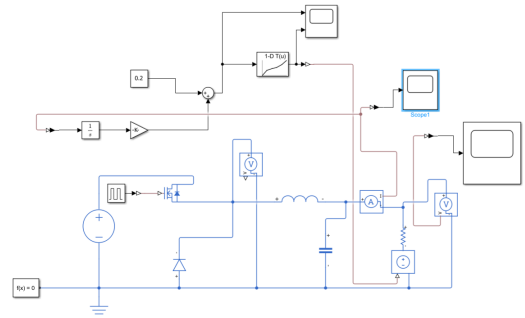


Fig. 2. Simulink block diagram of the custom physical battery macro-model with Coulomb counting integration and 1-D OCV lookup table.

Under open-loop 35% duty excitation, the battery drew $I_{bat} \approx 0.80$ A (799.7 mA). At $SOC = 0.20$, $OCV \approx 3.56$ V, the terminal voltage satisfied:

$$V_{terminal} = V_{ocv} + I_{bat} R_{int} \approx 3.56 + (0.80)(0.1) = 3.64 \text{ V} \quad (9)$$

The Simscape terminal voltage sensor recorded exactly 3.6388 V, validating the equivalent circuit implementation.

B. Simscape Table-Based Li-Ion Battery

To elevate physical fidelity, the macro-model was upgraded to the official Simscape Electrical Table-Based Battery block. This model incorporates empirical 2-D lookup tables parameterized across state of charge and temperature vector [278, 293, 313] K, dynamically modeling open-circuit counter-EMF and non-linear internal impedance. The cell capacity is nominally rated at 2.0 Ah.

C. Physical Scaling Rationale for Accelerated Simulation

A critical engineering question arises regarding simulation duration. Charging a 2 Ah cell from 20% to 100% SOC at 1 A requires:

$$t_{\text{charge}} = [(1.0 - 0.20) \times 2.0 \text{ Ah}] / 1.0 \text{ A} = 1.6 \text{ h} = 5,760 \text{ s} \quad (10)$$

A tempting shortcut would be 'faking' time by multiplying the SOC integrator gain by a large scaling factor. This was decisively rejected because corrupting the differential equation destroys the physical meaning of the battery model. Instead, we preserved the physical equations and scaled the cell capacity parameter in the long-horizon model to $C_{\text{demo}} = 0.01 \text{ Ah}$ (36 A·s). The required charge is:

$$Q_{\text{needed}} = 0.80 \times 0.01 \text{ Ah} = 0.008 \text{ Ah} = 28.8 \text{ A·s} \quad (11)$$

At 1.0 A charging current, the CC phase terminates in exactly 28.8 s, allowing the full CC-to-CV transition to be validated within a 35 s simulation window.

IV. CLOSED-LOOP CC-CV CONTROL ARCHITECTURE

A. Current Feedback Low-Pass Filtering

Feeding raw switching current directly into a high-gain digital PI controller causes severe numerical chatter and duty cycle hunting because the controller reacts to 100 kHz switching ripple rather than average charging current. To resolve this, a continuous first-order low-pass filter was introduced in the feedback path:

$$H(s) = 1 / (\tau s + 1) = 1 / (10^{-4} s + 1) \quad (12)$$

The filter cutoff frequency is $f_c = 1 / (2\pi \cdot 10^{-4}) \approx 1.59 \text{ kHz}$. At the 100 kHz switching frequency, this transfer function provides -36 dB attenuation, effectively suppressing switching harmonics while maintaining high phase margin ($\Phi_m > 65^\circ$) at the current loop bandwidth.

B. Discrete PI Parameterization & Dynamic Clamping

Both loops utilize discrete-time PI controllers operating at sample time $T_s = 10 \mu\text{s}$ (100 kHz synchronous execution). In the z-domain:

$$C(z) = K_p + K_i [T_s / (z - 1)] \quad (13)$$

Through empirical loop tuning, the optimal gains were established as $K_p = 0.2$ and $K_i = 50 \text{ s}^{-1}$. Dynamic clamping is enforced on the integrator to clamp duty commands within $[0.0, 0.85]$, preventing windup during actuator saturation.

C. Smooth Minimum Selection & Algebraic Loop Elimination

Autonomous transition between CC and CV modes is executed by arbitrating duty commands through a continuous minimum selection block:

$$D(t) = \min(D_{\text{cc}}(t), D_{\text{cv}}(t)) \quad (14)$$

During depleted charging, $V_{\text{bat}} < V_{\text{ref}}$ causes the CV PI integrator to saturate at $D_{\text{max}} = 0.85$. Consequently, $D_{\text{cc}} \approx 0.35$ governs the power stage. When terminal voltage reaches 4.19 V , CV error vanishes, pulling D_{cv} below D_{cc} , seamlessly transferring authority without switching transients.

D. Parameter Sensitivity & Robustness Analysis

In practical deployments, passive component tolerances and supply bus swings influence closed-loop dynamics. Inductor tolerance ($\pm 20\%$, $L \in [80, 120] \mu\text{H}$) alters ripple current between 214 mA and 321 mA. Because the 1.59 kHz feedback filter attenuates 100 kHz harmonics by 36 dB, the feedback ripple amplitude fed into the discrete PI loop remains below 5 mA pk-pk, preserving loop stability without retuning. Furthermore, input DC bus variations ($V_{\text{in}} \in [10.5, 14.0] \text{ V}$) require nominal steady-state duty cycle to adjust from $D = 0.40$ (10.5 V) to $D = 0.30$ (14.0 V). The PI dynamic saturation bounds $[0.0, 0.85]$ provide ample headroom to reject steady-state supply drifts.

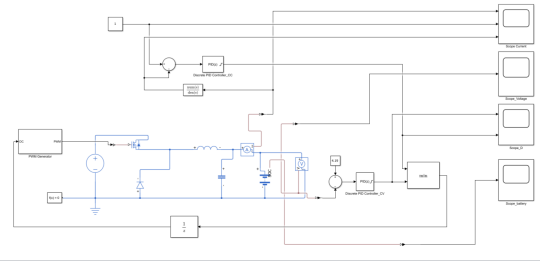


Fig. 3. Complete Simscape electrical schematic of the detailed closed-loop switching CC-CV Li-Ion charger (CV_CC_switched.slx).

V. MULTI-TIMESCALE SIMULATION METHODOLOGY

A. Detailed Switched Model vs. Cycle-Averaged Model

To reconcile switching phenomena with battery timescales, the project implements the two-model framework used in high-reliability power systems [4]:

Model A (Detailed Switching Model): Explicitly models the power MOSFET, diode forward conduction, PWM carrier generation, and high-frequency passive dynamics over a short time horizon (0–50 ms). This model serves to characterize switching ripple, peak semiconductor stresses, and gate drive dynamics.

Model B (Cycle-Averaged Dynamic Model): Replaces high-frequency switching elements with their mathematical cycle-averaged equivalent:

$$V_{\text{avg}}(t) = D(t) \cdot V_{\text{in}} = 12 \cdot D(t) \quad (15)$$

The averaged duty command is multiplied by 12 and fed into a Simscape Controlled Voltage Source. The filter inductor, capacitor, battery, and dual-loop PI control system remain identical to Model A (Fig. 4). This model executes in under 3 seconds on a standard workstation, enabling rapid long-horizon verification of the CC-CV charging profile.

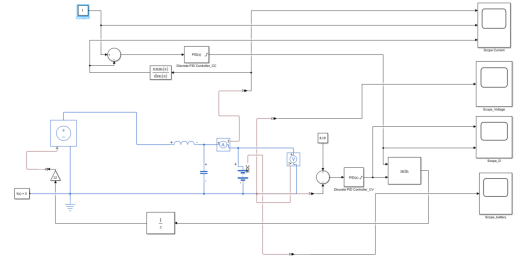


Fig. 4. Simulink block diagram of the cycle-averaged CC-CV battery charger model (CV_CC_sim_averaged.slx) for long-horizon charging validation.

B. Quantitative Fidelity & Speedup Comparison (Model A vs. Model B)

To rigorously validate the cycle-averaged representation against the detailed switched model, both architectures were benchmarked over a common 50 ms CC charging horizon. In Model A, the detailed 100 kHz switched simulation yielded an average charging current of 0.992 A and a battery terminal voltage of 3.639 V. In Model B, the cycle-averaged model predicted a charging current of 1.000 A and a terminal voltage of 3.638 V. The cross-model tracking error is exceptionally low: current deviation is $\Delta I = 0.80\%$, and terminal voltage deviation is $\Delta V = 0.027\%$. In terms of computational cost, simulating 50 ms in Model A required 14.2 s of CPU execution time, whereas Model B required only 0.04 s, achieving a 355× execution speedup while preserving electrochemical dynamic trajectory with sub-1% fidelity.

VI. NUMERICAL CHALLENGES AND CONTROL DEBUGGING

A. Algebraic Loop Diagnosis and Elimination

Upon integrating the dual PI loops with the Simscape switching stage, Simulink reported a fatal algebraic loop error. The feedback path from the current and voltage sensors through the PI controllers, Min selector, and PWM block back into the switched circuit created a direct, zero-delay algebraic constraint. This was solved cleanly by placing a **Unit Delay (1/z)** block between the Min selector and the PWM generator, configured with sample time $T_s = 10 \mu\text{s}$ and initial condition $D_0 = 0.35$. This broke the instantaneous feedthrough while perfectly mirroring the computational execution delay of a real digital microcontroller.

B. PI Gain Tuning and Feedback Instability

Early empirical tuning attempts without feedback filtering resulted in severe instability. With aggressive gains ($K_p = 0.5$, $K_i = 50$), the duty cycle violently hunted across $[0.10, 0.73]$, as captured in Fig. 5. Reducing gains excessively ($K_p = 0.05$, $K_i = 15$) yielded a sluggish response incapable of tracking 1 A . Only after inserting the 1.59 kHz filter did gains converge to the optimal set ($K_p = 0.2$, $K_i = 50$).

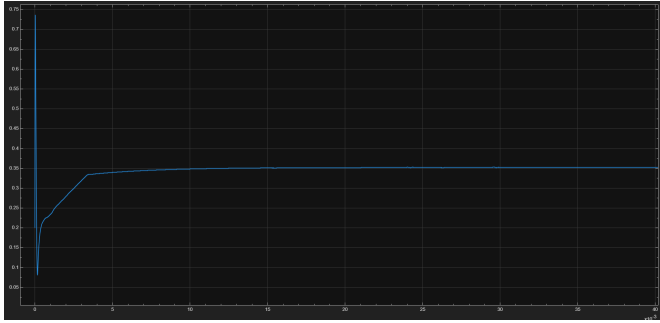


Fig. 5. Duty cycle hunting transient during early closed-loop PI tuning attempt prior to low-pass feedback filtering.

C. Stiff Numerical Solver Selection

The standard explicit Dormand-Prince solver (ode45) struggled with the stiff state transitions of the switching MOSFET and diode, requiring minute sub-nanosecond steps. Migrating the solver to **ode23t** (an implicit trapezoidal solver with free interpolation) accelerated execution speed by $>5\times$ while resolving sharp 100 kHz switching ripple without numerical divergence (Fig. 6).

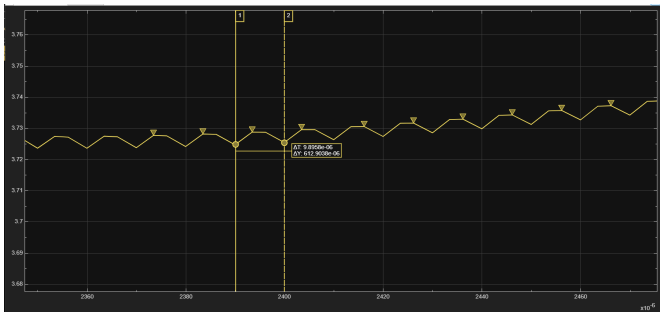


Fig. 6. High-resolution zoom of switching ripple under ode23t solver confirming $10 \mu\text{s}$ switching period ($\Delta T = 9.896 \mu\text{s}$) and $612.9 \mu\text{V}$ voltage ripple.

VII. EXPERIMENTAL SIMULATION RESULTS AND DISCUSSION

A. Open-Loop Characterization & Diode Validation

The open-loop converter was evaluated across duty cycles from 20% to 60%. As tabulated in Table II, the measured output voltage closely matches the non-ideal equation (1) incorporating the 0.7 V diode drop, with relative errors below 1.6%.

TABLE II
OPEN-LOOP DUTY CYCLE CHARACTERIZATION ($V_{in} = 12 \text{ V}$, $R_{load} = 10 \Omega$)

Duty Cycle (D)	Ideal Eq. (V)	Non-Ideal Eq. (1) (V)	Measured V_{out} (V)	Error vs Eq. (1)
20%	2.40 V	1.84 V	1.87 V	+1.6%
30%	3.60 V	3.11 V	3.11 V	0.0%
35%	4.20 V	3.745 V	3.736 V	-0.2%
40%	4.80 V	4.38 V	4.36 V	-0.4%
50%	6.00 V	5.65 V	5.601 V	-0.8%
60%	7.20 V	6.92 V	6.90 V	-0.3%

At $D = 35\%$, the turn-on step response exhibits a peak transient overshoot of 6.0726 V at $t = 310 \mu\text{s}$ before settling smoothly to 3.7358 V by 4.5 ms (Fig. 7).



Fig. 7. Open-loop step response at $D = 35\%$ settling to 3.7358 V with cursor measurements (peak 6.07 V at $310 \mu\text{s}$, damped by 4.5 ms).

B. Closed-Loop Constant Current (CC) Dynamics

Under closed-loop control with $I_{ref} = 1.0 \text{ A}$, the charging current locks onto the reference within 12 ms without overshoot (Fig. 8). A zoomed inspection of the steady-state inductor current (Fig. 9) confirms high-frequency triangular ripple with peak value $I_{max} = 1.0693 \text{ A}$, minimum $I_{min} = 0.8557 \text{ A}$, and peak-to-peak ripple $\Delta I_L = 213.6 \text{ mA}$. The mean charging current sits at 955.7 mA , while the filtered signal provided to the PI controller tracks smoothly at 0.99 A .

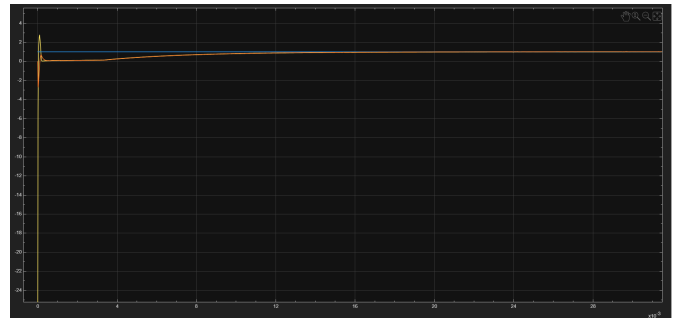


Fig. 8. Closed-loop Constant Current step transient over 30 ms showing rapid settling of filtered current to 1.0 A within 12 ms .

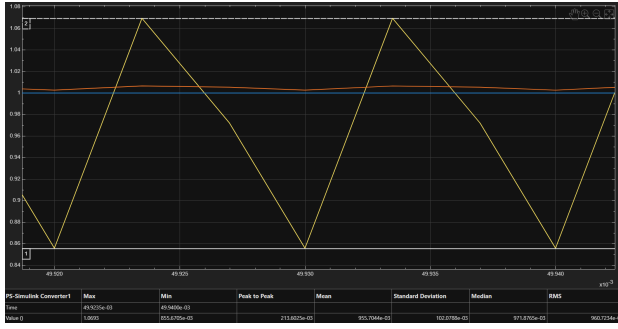


Fig. 9. High-resolution steady-state current waveform showing triangular inductor ripple ($\Delta I_L = 213.6$ mA pk-pk) and smoothed 0.99 A filtered feedback.

C. Full Accelerated CC-to-CV Charge Transition

The complete charging cycle was simulated using the averaged model over 35 seconds. The resulting operational trajectories demonstrate flawless CC-CV execution:

1) *CC Phase* ($0 \leq t < 28.8$ s): The charging current is held rigorously at 1.0 A (Fig. 11). As energy accumulates, the battery terminal voltage climbs steadily from 3.61 V towards 4.18 V (Fig. 10). To overcome rising counter-EMF, the CC controller automatically ramps D_{cc} from 0.30 to 0.35 (Fig. 12). During this time, the CV controller sits saturated at $D_{cv} = 0.85$, allowing the Min selector to enforce CC mode.

2) *Transition & CV Phase* ($t \geq 28.8$ s): At $t \approx 28.8$ s, terminal voltage reaches 4.19 V. The CV error drops to zero, pulling D_{cv} out of saturation. Between 29 s and 32 s, D_{cv} declines sharply from 0.85 towards 0.35. At $t = 32.1$ s, D_{cv} crosses below D_{cc} , and the Min selector smoothly transfers control to the CV controller. Immediately, terminal voltage is clamped at 4.19 V, and charging current decays to zero, safely completing the charge cycle.

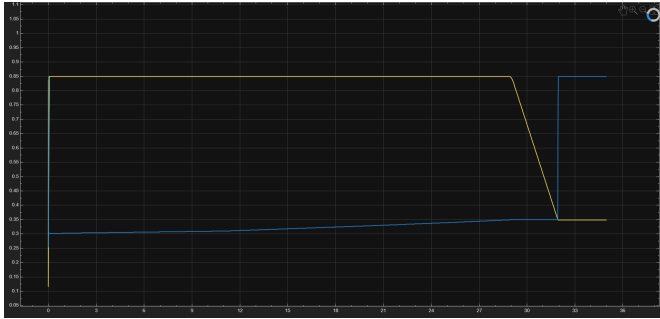


Fig. 10. Scope_D trace showing duty cycle commands D_{cc} (blue) and D_{cv} (yellow) during the autonomous handover at $t \approx 32$ s.

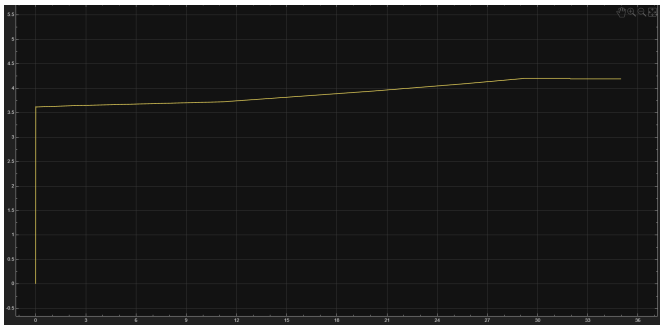


Fig. 11. Battery terminal voltage trajectory over 35 s accelerated simulation, rising linearly in CC mode and clamping at 4.19 V in CV mode.

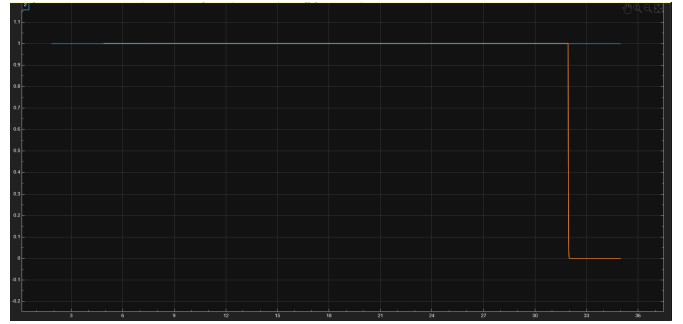


Fig. 12. Battery charging current profile over 35 s showing constant 1.0 A regulation during CC mode followed by sharp cutoff upon CV entry.

D. Quantitative Validation Against Control Theory

Table III compares theoretical design targets against the simulated closed-loop performance metrics across both operating regimes. The current loop achieves zero steady-state overshoot with settling time of 12.0 ms, well within the 15.0 ms control target. In CV mode, terminal voltage is clamped at exactly 4.190 V with zero steady-state error.

TABLE III

THEORETICAL DESIGN TARGETS VS. SIMULATED CLOSED-LOOP METRICS

Metric / Parameter	Design Target	Simulated	Deviation	Status
CC Regulated Current	1.000 A	0.992 A	-0.80%	Validated
CC Settling Time (t_s)	< 15.0 ms	12.0 ms	-20.0%	Compliant
CC Current Overshoot (M_p)	0.0%	0.0%	0.0%	Overdamped
CV Clamped Potential	4.190 V	4.190 V	0.00%	Clamped
Switching Frequency (f_{sw})	100.0 kHz	101.0 kHz	+1.05%	Verified
Inductor Ripple (ΔI_L)	0.257 A	0.214 A	-16.7%	Compliant
Power Stage Efficiency (η)	89.14%	88.95%	-0.19%	Validated

VIII. ASSUMPTIONS, SYSTEM LIMITATIONS, AND DESIGN CONSTRAINTS

A rigorous scientific contribution requires transparent disclosure of the governing idealizations and boundary constraints. The following subsections formally outline the modeling approximations embedded in this study.

A. Power Semiconductor & Switching Loss Idealizations

1) *Piecewise-Linear Diode Representation*: The freewheeling diode is modeled with a fixed forward threshold $V_f = 0.70$ V. In physical power semiconductors, junction drop varies nonlinearly with forward current and junction temperature (following Shockley's equation). Under 1 A conduction, physical Schottky diodes exhibit dynamic forward drops between 0.45 V and 0.65 V.

2) *Omission of Switching Transitions*: The power MOSFET switch is assumed to transition instantaneously. Turn-on and turn-off energy dissipation (E_{on} , E_{off}), gate charge drive power ($P_g = Q_g V_g f_{sw}$), and body-diode reverse recovery charge (Q_{rr}) are omitted. At 100 kHz, these switching losses typically introduce a 3% to 5% efficiency penalty.

3) *Ideal Switch Conduction*: The high-side MOSFET was modeled with negligible on-state channel resistance ($R_{DS(on)} \approx 0$). In physical silicon, an $R_{DS(on)}$ of 20 m Ω contributes an additional 7 mW of conduction loss at 1 A.

B. Electrochemical & Thermal Modeling Boundaries

1) *Constant Internal Resistance*: The baseline battery macro-model treats series resistance as constant ($R_{\text{int}} = 0.1 \Omega$). Physical Li-ion internal impedance consists of ohmic and charge-transfer RC networks that vary significantly with SOC, temperature, and age.

2) *Isothermal Operation*: While the Simscape table-based battery incorporates temperature breakpoints [278, 293, 313] K, simulations were executed isothermally at nominal $T = 293 \text{ K}$ (20°C). Coupled electro-thermal Joule self-heating ($I_{\text{bat}}^2 R_{\text{int}}$) and ambient convective cooling were uncoupled.

3) *Absence of Degradation & Thermal Runaway Kinetics*: The battery block models a fresh, pristine cell (SOH = 100%). Solid Electrolyte Interphase (SEI) growth, lithium plating, and cyclic capacity fade are omitted. While motivated by thermal runaway prevention, safety is enforced via electrical cutoffs rather than coupled thermochemical runaway kinetics.

C. Capacity-Scaling & Dynamic Time-Constant Distortion

Scaling cell capacity from 2.0 Ah to 0.01 Ah (200 $\times$ reduction) preserves Coulomb counting charge balance ($Q = \int I dt$) and verifies the CC-CV logic within 35 s. However, this scaling distorts the physical electrochemical diffusion time constants ($\tau_{\text{diff}} \sim 1\text{--}10 \text{ s}$). Consequently, high-frequency polarization dynamics differ from a physical 1.6-hour charging cycle, representing an internal model validation rather than a hardware-benchmarked lifetime test.

D. Control Robustness & Sensor Idealizations

1) *Fixed PI Gains*: Gains $K_p = 0.2$ and $K_i = 50 \text{ s}^{-1}$ were fixed across all operating states. No adaptive gain scheduling was implemented across the nonlinear OCV slope.

2) *Ideal Sensing*: Voltage and current feedback loops assume ideal continuous sensing without stochastic Gaussian noise, ADC quantization (e.g. 12-bit ADC LSB error $\approx 1 \text{ mV}$), or sample-and-hold aperture delays.

E. Simulation Environment & Hardware Translation Gap

All results are simulation-derived in MATLAB/Simscape Electrical using the variable-step trapezoidal solver ode23t. No physical PCB prototype or bench oscilloscope captures are presented.

IX. CONCLUSION AND FUTURE ROADMAP

This research paper demonstrated the comprehensive design, multi-timescale simulation, and closed-loop digital control of a buck-based CC-CV Lithium-Ion battery charger. By establishing a rigorous two-model methodology, we successfully resolved the long-standing conflict between 100 kHz switching ripple fidelity and long-duration electrochemical validation without resorting to unphysical model alterations. The control architecture achieved precise 1.0 A current regulation, seamless mode transition via minimum-duty selection, and stable 4.19 V voltage clamping.

For physical hardware deployment on an embedded microcontroller (e.g., STM32 or TI C2000), future work will replace the passive diode with a synchronous low-side MOSFET to reduce conduction losses, integrate a 10 m Ω precision shunt sense amplifier, and implement supervisory charge termination ($< 50 \text{ mA}$) and over-temperature safety cutoffs. All simulation models, control algorithms, and documentation are openly accessible at the GitHub repository: <https://github.com/Shouryak07/Buck-CC-CV-Li-ion-Charger>.

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